

Enterprise Digital Infrastructure

Basic concepts

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

1

Enterprise Digital Infrastructure

Enterprise digital infrastructure refers to the complex digital ecosystem used by companies and organizations to deliver public utilities and services, control industrial processes, manage businesses, ...

This ecosystem includes **technological** components:

✓ **hardware components**: servers, storage devices, backup devices, tape libraries, GPUs, FPGAs, sensors, actuators, ...

✓ **software components**: operating systems, DBMS, web applications, HPC applications, ...

✓ **communication/networking components**: wires, fibers, antennas, satellites, routers, access points, base stations, ...

The ecosystem also includes many **non-technological** components: power distribution systems, cooling systems, UPS, HVAC, smoke detectors, fire extinguishers, CCTVs, anti-intrusion systems, ...

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

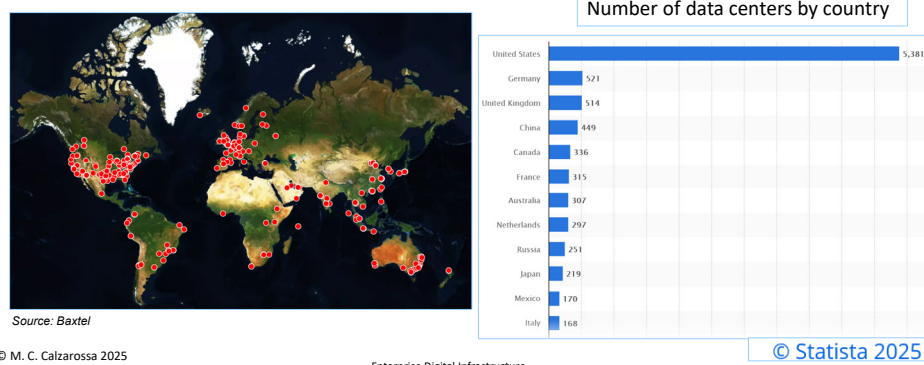
2

Data centers

Data centers are the “*smart buildings*” housing this infrastructure

They are the heart of the digital economy and represent the key in the supply chain of services and products and the backbone of our interconnected world, by facilitating the seamless flow of data

More than 10,000 data centers worldwide in 2024



3

Data centers (cont'd)

Data centers are owned and operated by companies/organizations for

- Running enterprise applications
- Deploying services
- Renting computing/storage resources
- Providing co-location services

Data centers are classified in *tiers* according to their characteristics [*standard by Uptime Institute*]

Public or private data centers

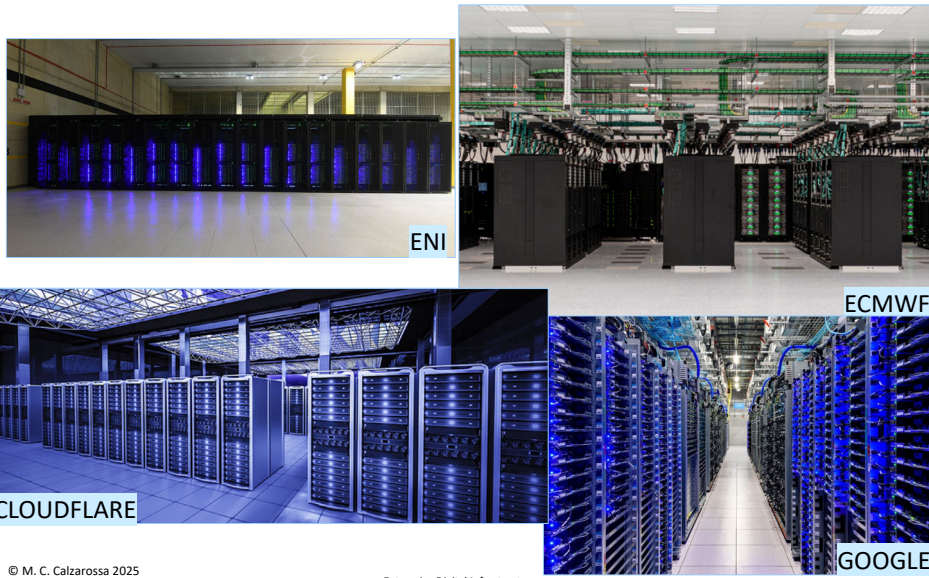
Services and computing/storage resources are accessible over the Internet

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

4

Inside data centers

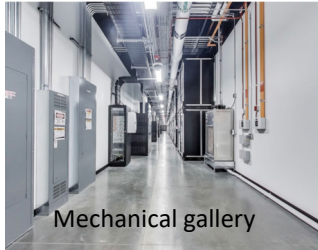


5

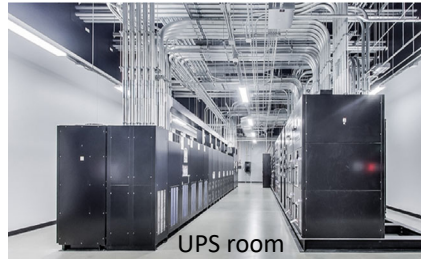
Inside data centers (cont'd)



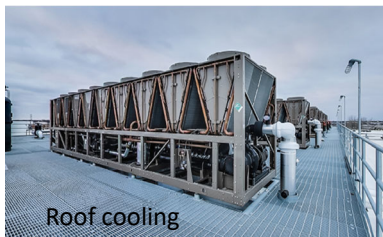
6



Mechanical gallery



UPS room



Roof cooling



Data hall

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

7

Data centers & energy

Energy-intensive nature of data centers

In the EU data centers used an *estimated* 45–65 Terawatt hours (TWh) of electricity in 2022, about 1.8–2.6% of total regional electricity consumption (25–30 TWh *estimated* for telecommunication networks)

Electricity consumption is growing rapidly as data centers take on more workloads and more energy-intensive workloads (e.g., artificial intelligence)

IDC expects global data center electricity consumption to more than double between 2023 and 2028 and reaching 857 TWh worldwide in 2028

Concerns about environmental impact (e.g., carbon footprint, water usage, waste production) and sustainability (e.g., energy efficiency)

Power Usage Effectiveness (PUE) measures energy efficiency as ratio between power entering data center and power used to run digital infrastructure (smaller is better)

Solar and wind power offer significant environmental advantages and low cost

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

8

Data centers & renewal energy sources

Renewal energy sources for carbon neutral data centers



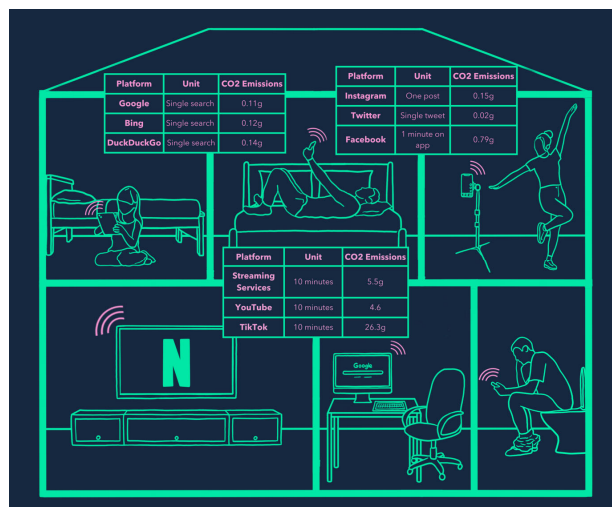
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

9

Emissions from digital technologies

In the EU, greenhouse gas emissions from digital technologies are comparable to emissions of aviation sector



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

10

Recap

Definition of enterprise digital infrastructure

Data centers and energy issues

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

11

Choice of the infrastructure

The choice of the “right” technological components is very challenging:

- ingredients vs recipe
- intuition

The choice depends on:

- ✓ type of service/application
- ✓ workload/traffic
- ✓ quality/performance requirements
- ✓ cost

Two contradictory requirements:

- | | | |
|-----------|-------------------|------------------------------------|
| – quality | → efficiency | → <i>over provisioning</i> |
| – cost | → competitiveness | → <i>under provisioning</i> |

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

12

Performance challenges

Services are *on-demand* (utility computing)

Service *popularity* varies with time due to their social nature and to external events

- Network effect
- Flash crowd
- Slashdot effect

Location-based/context-aware services use location data to customize the content

Services deployed on *geographically distributed* (virtualized) infrastructure

Composition of multiple services dynamically selected and of heterogeneous data flows under the control of different providers

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

13

Functional vs non-functional requirements

Functional requirements define what a system/service is supposed to do, that is, its **technical properties**

Non-functional requirements define how a system/service should behave and the **experience** of the users

No direct relationship

A system can still work if non-functional requirements are not met, although it might not satisfy user or stakeholder expectations or the business needs

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

14

Quality of Service (QoS)

Quality is a very important requirement for an effective digital transformation

Quality of Service (QoS) refers to *non-functional requirements* related to service behavior and user experience (satisfaction/dissatisfaction)

- Dependability
 - Reliability (uptime and downtime)
 - Trustworthiness (privacy and security)
- Performance
- Usability

Resilience refers to the ability to maintain essential operational capabilities in the face of adversity (e.g., flash crowd, security attack, power outage) and rapidly recover to an effective operational posture [Source: NIST]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

15

QoS attributes

Very *subjective* attributes: perception of the users depends on many factors, e.g., their type/skill/age, service type, level of interactivity, timing requirements

Concept of *patience*: ability to accept delays without getting upset



How long users can wait? Some numbers: 0.1 sec “instantaneous” response, 1 sec still ok, 10 sec so-so, >10 sec too much

Generation Z is less patient than older people



Patience loss leads to **frustration**

Lack/loss of patience significantly affects online businesses, e.g., loss of customers, drop in sales/revenues, reputation damages

Waiting times should be taken into account on the design phase

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

16

Usability

Broad quality attribute that includes many aspects of a service

It measures the experience of the users in interacting with systems/services (*user satisfaction*)

It is defined as the ability to **learn** and **use** a given system/service and **memorize** it

Very subjective

It is expressed in terms of



- *efficiency*: ability to complete a task easily, quickly and without frustration
- *effectiveness*: ability to successfully complete a task (and obtain the desired results)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

17

Usability (cont'd)

The usability depends on

- the design of the user interface (UI)
- the technological infrastructure and its performance

Some services (e.g., video/audio streaming, VoIP) might become unusable because of poor performance of the technological infrastructure on the provider side (but also on the user side)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

18

Quality of Service (QoS)

Quality of Service (QoS) attributes

Resilience is the persistence of dependability when facing changes

- Dependability
 - Reliability (uptime and downtime)
 - Trustworthiness (privacy and security)
- Performance
- Usability

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

19

Reliability

Reliability is the ability of a given (hardware or software) component to perform/work as planned (i.e., continuity of correct behavior)

Reliability includes the concept of *availability*

Availability is the quality of a system/service of being reachable and working properly whenever requested (i.e., readiness)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

20

Availability

Availability defined as the fraction of time during which a system/service is reachable and works properly

$$\text{availability} = \text{uptime} / (\text{uptime} + \text{downtime})$$

The ideal case 24/7 (=100%) availability is desirable but unrealistic

Downtimes noticed immediately (we are addicted)

The tolerance of the users to unavailability is very *subjective*

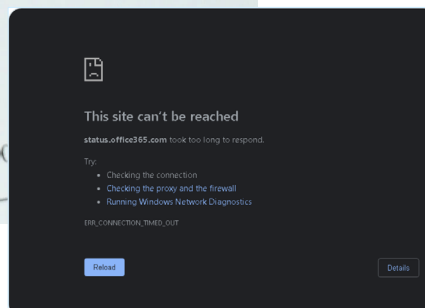
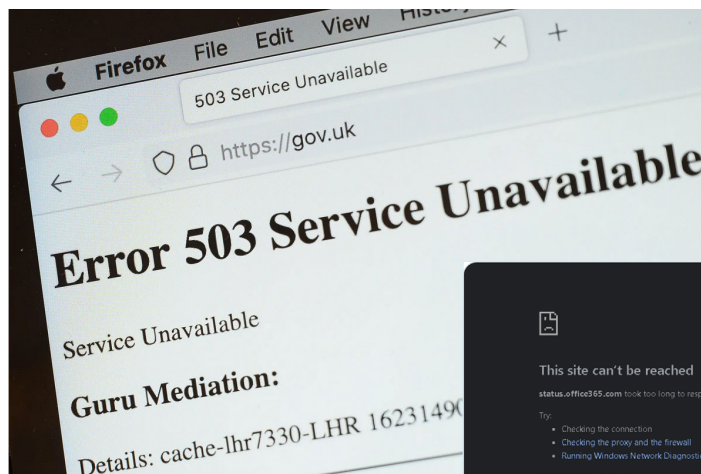
Unavailability may cause many inconveniences, inefficiencies, damages to reputation, loss of trust, financial losses, ...



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

21



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

22

Tech companies outages in the news

Facebook, Instagram and other Meta apps go down due to 'technical issue'

PUBLISHED WED, DEC 11 2024-2:38 PM EST | UPDATED WED, DEC 11 2024-6:38 PM EST

Meta says it has almost resolved outage on Facebook and Instagram apps

By Ramtishah Maruf, CNN
© 2 minute read · Updated 5:45 PM EST, Wed December 11, 2024

PlayStation Network experiences massive outage, frustrating gamers nationwide

PUBLISHED SAT, FEB 9 2025-2:52 PM EST

PlayStation Network outage leads to rush of complaints from gamers

Problems began in UK at about midnight on Friday, with more than 71,000 users complaining of losing access to online events

Sony's PlayStation Network reports major global outage

By Azeemah Bacon, CNN
© 2 minute read · Updated 7:39 PM EST, Sat February 8, 2025

Sony Reveals Compensation For All PS Plus Players After PSN Outage

By Paul Tassi, Senior Contributor, News and opinion about video games, televisio...
Feb 10, 2025, 09:20am EST

PSN Is Still Down After 24 Hours And No One Knows Why (Update: It's Back)

By Paul Tassi, Senior Contributor, News and opinion about video games, televisio...
Feb 08, 2025, 03:44pm EST

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

23

Largest IT outage (until now)

BBC

Mass IT outage affects airlines, hospitals, media and banks

19 July 2024

Willkommen im Check-in 1

Airline	Flight	Status	Gate
Austrian	OS 309	On Time	10
British Airways	BA 149	On Time	11
easyJet	U2 8555	On Time	12
Lufthansa	LH 459	On Time	13
Swire	SC 33	On Time	14

S.No	Flight	Destination	STD	Gate
1.	UK 62	YOR	13:25	53
2.	UK 75	YOR	13:15	52
3.	UK 77	AIR	12:45	52
4.	UK 47	DCA	13:50	60
5.	UK 69	ATO	12:55	49
6.	UK 61	2XO	"	39-A
7.	UK 10	Yardgoat	13:25	39-B
8.	UK 5154	MAA	12:30	39A
9.	UK 5162	RAI	13:25	56
10.	UK 5262	SXR	13:45	58
11.	UK 5204	YAH	13:45	49

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

The New York Times

Chaos and Confusion: Tech Outage Causes Disruptions Worldwide

Airlines, hospitals and people's computers were affected after CrowdStrike, a cybersecurity company, sent out a flawed software update.

Global IT chaos persists as CrowdStrike boss admits outage could take time to fix

19 July 2024



24

Outage impacts



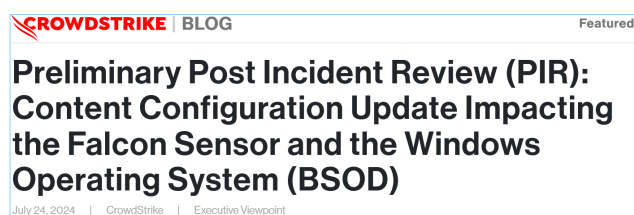
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

25

What actually happened?

CrowdStrike released a preliminary post incident review a few days after the incident was detected



About a month later they released another report on [Root Cause Analysis](#)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

26

Financial loss

Table 2: Fortune 500 Financial Loss by Industry

Industry Sector	Annual Revenue, All Companies	Annual Revenue, Impacted Companies	Estimated Direct Financial Loss	Estimated Direct Financial Loss, per Company
Health	\$2.77T	\$2.24T	\$1.94B	\$64.60M
Banking	\$0.82T	\$0.74T	\$1.15B	\$71.84M
Transportation - Airlines	\$0.19T	\$0.19T	\$0.86B	\$143.38M
Software & IT related Services	\$1.71T	\$0.89T	\$0.56B	\$112.59M
Retail/Wholesale	\$2.89T	\$0.73T	\$0.47B	\$14.73M
Other	\$5.15T	\$0.95T	\$0.19B	\$12.60M
Finance	\$0.68T	\$0.17T	\$0.14B	\$17.21M
Transportation - Other	\$0.36T	\$0.26T	\$0.07B	\$11.10M
Manufacturing	\$3.37T	\$0.31T	\$0.04B	\$5.93M
Total	\$17.94T	\$6.47T	\$5.41B	\$43.64M

Source: Parametrix Impact Analysis: CrowdStrike's Impact on the Fortune 500

GeoTech Cues | August 6, 2024

The Great IT Outage of 2024 is a wake-up call about digital public infrastructure

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Another outage

McDonald's suffers global tech outage, shutting some restaurants for hours

McDonald's apologizes for global system outage that shut down some stores for hours

McDonald's blames global outage on third party

March 15 Status:

At approximately midnight CDT on Friday, McDonald's experienced a global technology system outage, which was quickly identified and corrected. Many markets are back online, and the rest are in the process of coming back online. We are closely working with those markets that are still experiencing issues. Notably, this issue was not directly caused by a cybersecurity event; rather, it was caused by a third-party provider during a configuration change.

Reliability and stability of our technology are a priority, and I know how frustrating it can be when there are outages. I understand that this impacts you, your restaurant teams and our customers. What happened today has been an exception to the norm, and we are working with absolute urgency to resolve it. Thank you for your patience, and we sincerely apologize for any inconvenience this has caused.

Brian Rice
EVP, Global Chief Information Officer

McDonald's Tech Outage: A Wake-Up Call for Enhanced Digital Resilience

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

And another ...

ChatGPT down: OpenAI working to fix major global outage after spike in reported problems

ChatGPT Faces Global Outage, Users Flood Social Media With Complaints

The complaints indicate that the issue is widespread, with reports coming in from different countries.

ChatGPT Down: Internet in uproar as users panic over AI dependency

ChatGPT back online after outage which hit thousands worldwide

23 January 2025

The New York Times

ChatGPT Is Restored After Hours-long Outage

Thousands of users were unable to access OpenAI's chatbot starting around 6 a.m. on Thursday. Shortly after 10 a.m., the company said the issue had been resolved.

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

29



ChatGPT ?



Increased errors for ChatGPT

Incident Report for OpenAI

Resolved	This issue has now been resolved. Between 4:23am and 7:10am PST, customers experienced elevated errors on ChatGPT Posted 1 month ago. Jan 23, 2025 - 08:05 PST
Update	We are continuing to monitor for any further issues. Posted 1 month ago. Jan 23, 2025 - 07:11 PST
Monitoring	A fix has been implemented and we are monitoring the results. Posted 1 month ago. Jan 23, 2025 - 07:09 PST
Update	We are continuing to work on a fix for this issue. Posted 1 month ago. Jan 23, 2025 - 06:34 PST
Identified	We have identified the root cause of this issue, and are currently working to implement a fix. Posted 1 month ago. Jan 23, 2025 - 05:43 PST
Investigating	We are currently experiencing elevated error rates for ChatGPT. We are currently investigating. Posted 1 month ago. Jan 23, 2025 - 05:12 PST

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

30

OpenAI

Postmortem

Impact

Between 3:30am PT and 7:08am PT [chatgpt.com](#) and a small number of APIs experienced elevated error rates.

Root Cause & Remediation

The root cause of the initial issue was a failure with Cosmos DB in ChatGPT. Our service provider recovered the affected DBs at 4:25am PT.

The second issue was caused by web service pods crash-looping due to their health checks failing, marking pods as unhealthy and subsequently not having enough resources available to service all the requests to the web layer. At this time we believed this was due to users retrying requests. These issues recovered after we implemented a fix to the web server's health checking logic.

Prevention

1. We will make a series of changes to the web systems to more gracefully degrade when they fail including concurrency control and better load shedding.
2. A larger investment in making our DB layer more resilient to service provider outages is already underway.

We know that extended outages can impact your products, businesses, and lives. We will prioritize the preventative measures outlined above to continue improving our reliability.

Posted 26 days ago, Jan 30, 2025 - 07:58 PST

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

31



How we define availability

WhatsApp Business Platform is considered available unless it has been unable to send or receive messages for at least 10 consecutive minutes.

How we define latency

Latency measures how long it takes to send one-to-one inbound and outbound messages, starting when a message is received by the WhatsApp Business Platform and ending when it is ready for pickup by the business or the client.

Availability

99.94% last 31 days

31 days ago

1 days ago

Latency

1187 ms (P90), 2303 ms (P99) last 31 days

Cloud API

Resolved

Resolved: Cloud API

Feb 28 2025 5:12 PM GMT+1

We have recovered from an earlier outage impacting Cloud API have now been restored. We apologize for any inconvenience that this may have caused.

High disruptions: Cloud API

Feb 28 2025 4:56 PM GMT+1

We are currently experiencing high disruptions impacting sending messages on WhatsApp Cloud API. This issue started at 28/2/2025 3:56 PM GMT. Our engineering teams are investigating the issue. We will provide another update within 1 hour or sooner if additional information is available.

© M. C. Calzarossa 2025

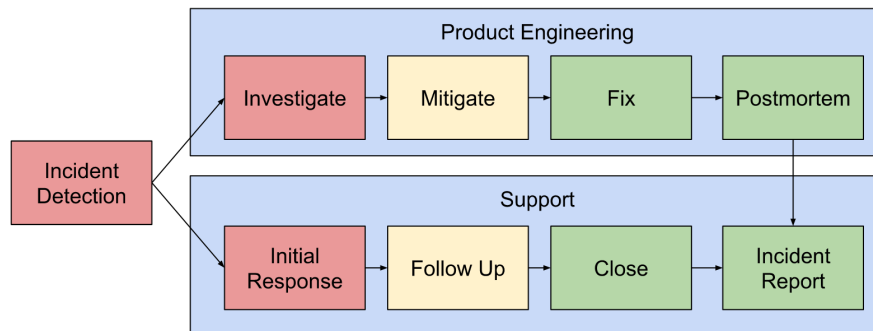
Enterprise Digital Infrastructure

32

© University of Pavia 2025

How Google manages incidents

Two parallel workflows each consisting of multiple steps



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

33

Scheduled downtimes

Scheduled downtimes refer to *planned* downtimes for system maintenance/upgrade



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

34

Scheduled maintenance

Scheduled maintenance

Scheduled for Feb 26, 2025 01:00-02:00 UTC

Dropbox will be performing routine maintenance on February 25, 2025 starting at 5:00 PM (Pacific). You may not be able to access Dropbox for a short period of time, but we'll have things back up and running as soon as possible to minimize disruption to you.

Posted on Feb 24, 2025 - 16:07 UTC

Announcement

Aries Systems will undergo a maintenance period for Editorial Manager® and ProduXion Manager® on Monday, March 3 starting at 3PM EST and ending at 7PM EST. Users may experience slowness or connectivity issues during this time. Please ensure to save your work before the maintenance period begins. We appreciate your patience and apologize for any inconvenience.

Change #	Date Posted	Location	Sector	Webex Service	Description
CHG0682463	Feb 25, 2025, 23:32 UTC	Frankfurt, Amsterdam, London, and Manchester Data Centers	Commercial	Webex Calling	Webex Calling Service Maintenance: March 11, 2025



Scheduled Maintenance on February, 24

UPD: The maintenance is now completed.

Telegram underwent planned maintenance between 2:00AM and 3:30AM UTC on February, 24.

We did our best to ensure that Telegram remained fully operational around the world throughout the entire maintenance window.

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Recap

Quality of Service is a very visible non-functional requirement for the services and applications deployed nowadays

Very subjective

Quality assessment is based on attributes referring to *measurable* properties

- *Usability*: the technological infrastructure has to match the *characteristics* of the load and the *intensity* of traffic it is supposed to handle
- *Availability*: a service must be *ready* whenever users request it

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

36

Outage analysis

Avoiding digital infrastructure failures and downtime is of paramount importance

According to Uptime Institute, the overall frequency and severity of outages is decreasing [[Uptime Institute 2024 Outage Analysis](#)]

Outage information is commercially sensitive and subject to uncertainty

Outage are costly in terms of financial loss, business and customer disruption and reputational loss

More than half of the outages cost more than 100,000 USD

Outages are mainly caused by:

- Failures of non-technological components (power, cooling)
- Human errors & management errors (ignored/inadequate procedures)
- Software, system & network errors (bugs, misconfigurations)
- Third-party issues (cloud, telco providers)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

37

Availability: Lessons learnt

Downtimes are costly and must be minimized (*pit stop*)

Unscheduled downtimes (due to power outage, software bugs, security attacks, misconfigurations, sudden traffic surge, ...) must be managed very carefully with timely announcements, regular follow-ups, mitigation and workaround actions, detailed incident reports ... *information & transparency*

Necessary to *prevent* events that cause unscheduled downtimes

Scheduled downtimes (due to hardware/software maintenance, installation, configuration, upgrade, ...) must be planned very carefully with timely *announcements* and development of *mitigation plans* whenever possible

The availability depends on human errors and external events but also on the *reliability* of the technological infrastructure

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

38

Reliability

Reliability: probability of a hardware/software component to work as planned for a specified time interval (i.e., continuity)

Fault → *Error* → *Failure* chain

- *Fault*: defect inside a system that causes of an error, e.g., software bug
- *Error*: deviation of the system from the correct service state, e.g., software bug that manifests itself
- *Failure*: event that occurs when a system deviates from its correct behavior and stops performing the required function, e.g., software bug makes the system fail

A reliable component seldom fails

Components should be *robust* or *fault-tolerant* (i.e., able to mask faults)

© M. C. Calzarossa 2025

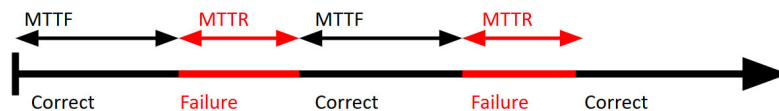
Enterprise Digital Infrastructure

39

Reliability (cont'd)

Robustness of a component (e.g., storage device) is measured in terms of MTTF (or MTBF or Annualized Failure Rate)

Recovery time of a failed component is measured in terms of MTTR



It is necessary to deploy specific *redundancies*

- *Replication* of hardware components
- *Diversity/distribution* of hardware and software components

Examples: power supplies, UPS, RAID architectures, providers

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

40

Reliability: Google's perspective

Google's application and network architecture is designed for **maximum reliability and uptime**. Google's computing platform assumes ongoing hardware failure, and it uses robust software failover to withstand disruption. All Google systems are inherently **redundant by design**, and each subsystem is not dependent on any particular physical or logical server for ongoing operation. Data is **replicated multiple times** across Google's clustered active servers so that, in the case of a machine failure, data will still be accessible through another system. We also replicate data to secondary data centers to ensure protection from data center failures.

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

41

Data center reliability

Tier classification system is a proprietary standard created more than 25 years ago to *assess* and certify the reliability of data centers [*Uptime Institute*]

The classification explains the infrastructure required for data center operations

There are different Tiers according to the system availability needed

Tiers match a particular business function and define criteria for maintenance, power, cooling and fault capabilities

Four progressive Tiers: each Tier incorporates the requirements of the lower Tiers

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

42

Data center Tiers

Tier I: simple warehouse with area for IT systems, engine generator, UPS & cooling system; shutdown required for preventive maintenance and repairs; severe consequences from failures

Tier II: partial redundancy with some redundant components for power and cooling

Tier III: “concurrently maintainable” with N+1 availability (Need plus 1); redundant distribution paths; no shutdowns for maintenance or replacement

Tier IV: 2N (Need times 2) redundancy and a 99.99% uptime guarantee, without any single point of failure; fault-tolerant and occurrence of any unplanned event while still maintaining operations

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

43

Reliability : Lesson learnt

Service continuity is ensured at the expense of an increased complexity (and cost) of the infrastructure

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

44

Trustworthiness

Trustworthiness (i.e., worth of trust/confidence) is the ability of a system to *protect* data, devices and infrastructures

This concept includes *security* and *data privacy*

Confidentiality, Integrity & Availability (CIA triad) of information is a common model used to develop security systems

Many types of security threats and attacks:

- DDoS
- data leakage/breach
- malware
- phishing
- ransomware
- ...

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

45

Denial of Service attacks

DDoS-ing is a crime

A Denial of Service (DoS) is a malicious targeted cyberattack whose goal is to take the “victim” (system or service) down and make it *unavailable* to its legitimate users



A DDoS, i.e., Distributed DoS attack, is a subclass of DoS attacks involving multiple devices, i.e., a **botnet**, to make attack stronger and more effective

These attacks are popular “weapons” for diverse threat actors (geopolitical motivated hackers, cyber vandals) since they are highly noticeable and impact an entire online user base

Their objectives are to cause damages, productivity loss and financial losses and to gain public attention

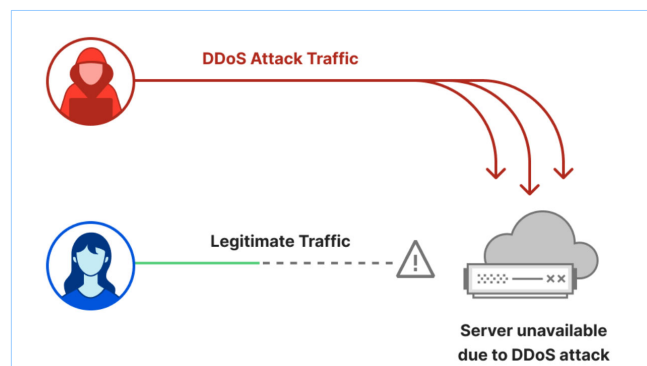
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

46

Denial of Service attacks (cont'd)

Attacks are implemented by overloading the target with illegal load with the objective of exhausting the target resources (e.g., bandwidth, processing power, memory)



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

47

Botnets

Botnet = robot + network: an army of comprised “zombie” devices injected with virus or malware and controlled from a remote location frequently without knowledge and consent of the owner

IoT devices (routers, home appliances) are often unmanaged & unmonitored, thus excellent targets of cyber-criminals for creating botnets (*thingbot*)

Devices infiltrated using spam/phishing emails, brute force attacks, misconfigurations, exploitation of [Common Vulnerabilities and Exposures](#) (CVE)

Botnets can be created (building kits, tutorials for Building-Your-Own-Botnet) or rented for very little money

Botnets are ILLEGAL!!!

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

48

DDoS & botnets

A DDoS-for-hire is an *on demand* service powered by botnets consisting of two main components:

- Stresser, i.e., a software used to “stress-test” the resilience of a system
- Booter, i.e., the illegitimate use of a stresser

No need of (any) technical know-how to launch attacks

Affordable services available on the dark web and even on the web

\$23.99 1 month	\$34.99 1 month	\$44.99 10 years
1 Month Gold	1 Month Diamond	Lifetime Bronze
Time per boot2400 sec	Time per boot3600 sec	Time per boot600 sec
Concurrents1	Concurrents2	Concurrents2
Total network220Gbps	Total network220Gbps	Total network220Gbps
ToolsIncluded	ToolsIncluded	ToolsIncluded
Support24/7	Support24/7	Support24/7
Buy with Paypal 	Buy with Paypal 	Buy with Paypal 
 bitcoin	 bitcoin	 bitcoin

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

49

Police from 15 Countries Shut Down Major DDoS-for-Hire Platforms



Vlad CONSTANTINESCU
December 12, 2024

Promo Protect all your devices, without slowing them down.
[Free 30-day trial](#)



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

50

EUROPOL

Law enforcement shuts down 27 DDoS booters ahead of annual Christmas attacks

Operation PowerOFF is an ongoing multifaceted operation against DDoS-for-hire services, designed to hold providers and users accountable

NCA infiltrates world's most prolific DDoS-for-hire service

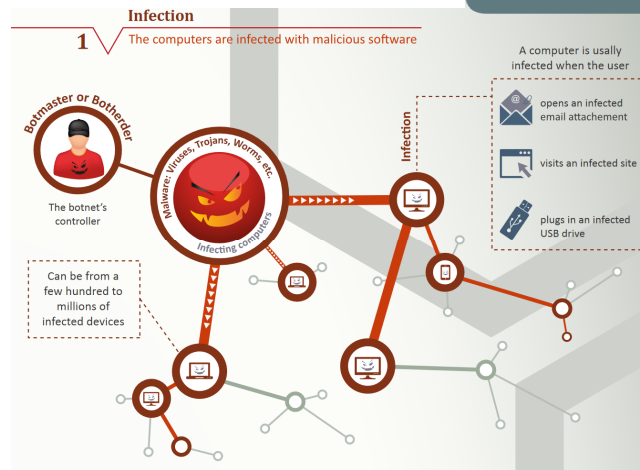


© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

51

Infection phase

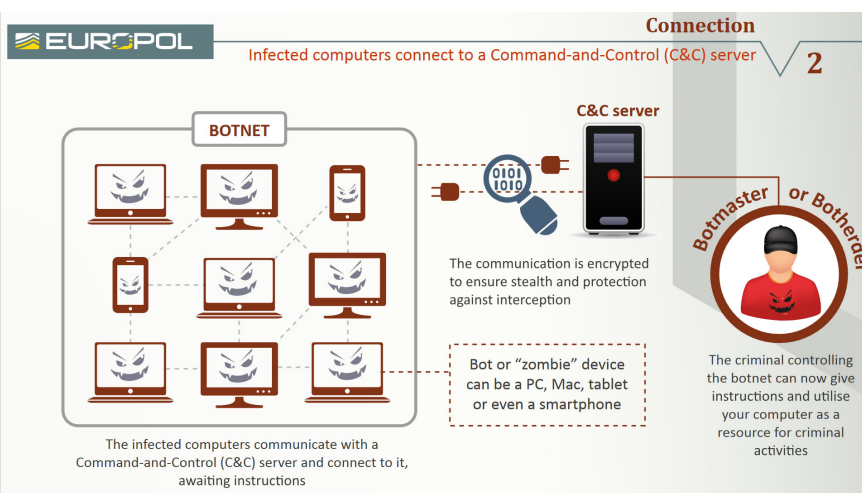


© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

52

Connection phase

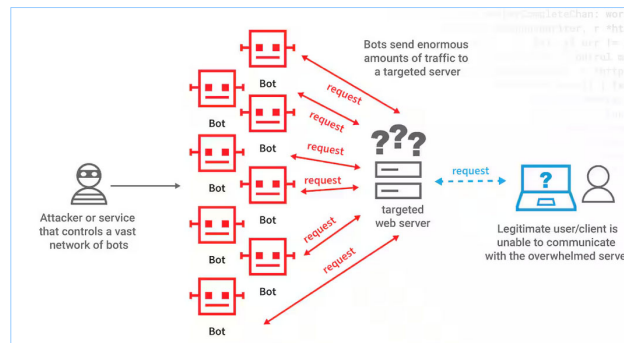


© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

53

DDoS implementation



The larger the botnet the more disruptive the attack is

Mirai botnet is a very large historical botnet (first appeared in 2016) consisting of compromised IoT devices (e.g., DVRs, IP cameras) running on ARC processor and infected through a login with default credentials

Mirai has been responsible of many DDoS attacks

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

54

Types of DDoS attacks

DDoS attacks can be classified in various categories depending on their goal

1. *Volumetric* or *flood-based* attacks designed to consume network bandwidth (implemented at the network, transport or application layers)
2. *Computational* attacks designed to consume CPU and memory (usually implemented at the application layer via TCP)
3. *Vulnerability-based* attacks exploiting software vulnerabilities (HTTP/2 Rapid Reset "request, cancel, request, cancel" pattern)

Attacks can also be classified as *symmetric* or *asymmetric* depending on the amount of resources required from the attackers

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

55

Famous DDoS attack: Dyn

Massive cyberattack turned ordinary devices into weapons

by Samuel Burke @CNNTech

October 22, 2016: 10:37 AM ET



DDoS attack that disrupted internet was largest of its kind in history, experts say

Dyn, the victim of last week's denial of service attack, said it was orchestrated using a weapon called the Mirai botnet as the 'primary source of malicious attack'

● **Major cyber attack disrupts internet service across Europe and US**

Peak traffic 1Tbps

DDoS attack on Dyn came from 100,000 infected devices

DNS service provider Dyn says Mirai-powered botnets were the primary source for Friday's disruption



By Michael Kan

U.S. Correspondent, IDG News Service | OCT 26, 2016 2:21 PM PDT

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

56

Other famous DDoS attacks

LILLY HAY, NEWMAN SECURITY 03.01.18 11:01 AM

GITHUB SURVIVED THE BIGGEST DDOS ATTACK EVER RECORDED

New world record DDoS attack hits 1.7Tbps days after landmark GitHub outage

Memcached denial-of-service attacks are getting bigger by the day, according to new analysis.

By Liam Tung | March 6, 2018 -- 12:34 GMT (12:34 GMT) | Topic: Security

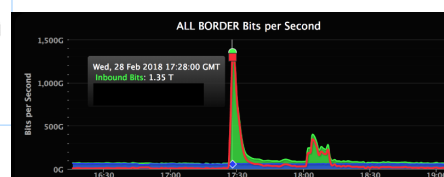
DDoS attacks exploited misconfigured *memcached* servers

Memcached is a high-performance open source distributed memory object caching technology designed to speedup dynamic web applications by alleviating database load

Amplification/reflection DDoS attacks coming from UDP port 11211 (i.e., spoofed requests to vulnerable servers that respond with a much larger amount of data, 50K amplification factor)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure



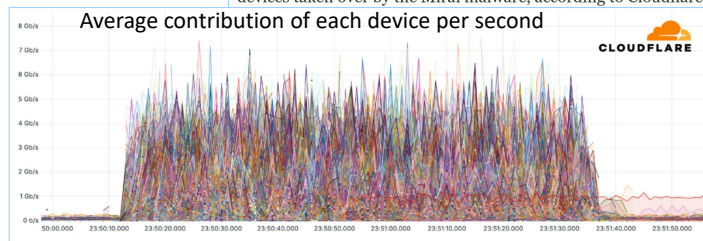
57

Record-Breaking DDoS Cyberattack Reaches 5.6 Tbps, Cloudflare Reports

By [Kate O'Flaherty](#), Senior Contributor.
Jan 22, 2025, 08:38am EST

Cloudflare has mitigated the biggest distributed denial of service (DDoS) attack on record, according to a new report.

The DDoS attack, which hit an East Asia based telecoms firm last year, reached 5.6 terabits per second (Tbps) and lasted for 80 seconds. It was comprised from a botnet of 13,000 devices taken over by the Mirai malware, according to Cloudflare's [blog](#).



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

58

Akamai Prevents the Largest DDoS Attack on a U.S. Financial Company

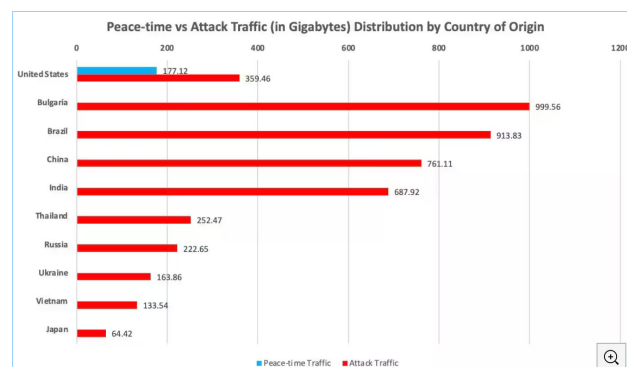


Craig Sparling & Sandeep Rath
September 08, 2023

Attack based on a combination of ACK, PUSH, RESET, and SYN flood attack vectors (peak: 633.7 Gbps and 55.1 Mpps)

It lasted two minutes

Cumulative traffic over 24 hours



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

59

DDoS summary

Impacts of DDoS:

- Unavailability of systems/services
- Financial losses
- Reputation loss

Mitigation actions depend on the type of systems/services under attack

- Disable UDP
- Prevent IP spoofing
- Use sophisticated (expensive) technologies

© M. C. Calzarossa 2025

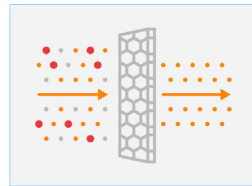
Enterprise Digital Infrastructure

60

Scrubbing centers

A centralized data cleansing solution aimed at mitigating DDoS attacks

Network traffic is inspected and proactive or custom mitigation controls designed to detect and stop attacks instantly are applied



Akamai provides a network of scrubbing centers

Traffic is usually redirected (using DNS or BGP) to the closest scrubbing center where potential attacks are mitigated (e.g., by blocking malicious IP addresses) and clean traffic is sent to the network for delivery

Scrubbing centers are equipped to sustain high volumetric floods (up to 20 Tbps)

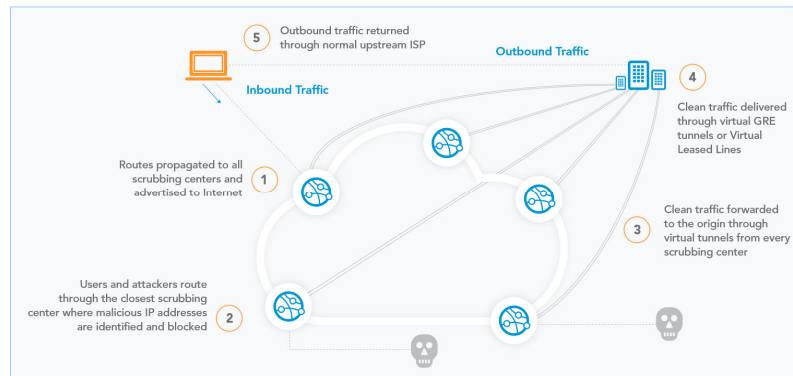
Often used by big enterprises, ISPs, cloud providers

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

61

Akamai scrubbing architecture



Source: Akamai

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

62

Data leakage/breach



Data leakage/breach refers to the *disclosure of sensitive data*

Three main categories of sensitive data that are target of cyber-criminals:

- *Personally Identifiable Information (PII)* refers to data (e.g., name, home address, phone number, credit cards, health records) whose disclosure allows the identification of specific individuals (e.g., *identity theft*)
- *Business sensitive data* refer to data (e.g., intellectual property, designs, patents, financial data, customer/supplier info) whose disclosure to competitors poses a company at risk (e.g., *extortion*)
- *Classified information* refers to data (e.g., Wikileaks, Snowden & NSA) whose disclosure poses governments/countries at risk (e.g., *terrorism*)

Intentional data leakage: data captured as a consequence of a cyber attack

Unintended data leakage: data disclosed to strangers by data “owners”

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

63

Recap

Reliability refers to continuity of services and applications

It is necessary to implement appropriate redundancies of technological components and data

Trustworthiness is another aspect of dependability referring to the concepts of security and privacy

DDoS is a serious security attack that exploits botnets to take down system and services

Appropriate measures have to be implemented to mitigate attacks

Data leakage/breach refers to intentional or unintended disclosure of sensitive data

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

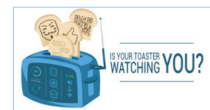
64

Unintended data leakage

General tendency to oversharing online and overlooking the implications and security risks

Social media decrease the *privacy* of individuals

- Apps with permissions to access data on our devices (**with our consent**)
- Location data associated with posts, tweets, photos, ...



Social engineering techniques used to persuade individuals to disclose personal data by stimulating their curiosity and taking advantage of their inexperience and gullibility (e.g., phishing)



Loss/theft of mobile devices or removal media

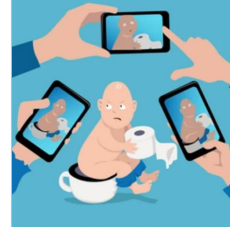


© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

65

Oversharing



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Unintended data leakage: what to do

Importance of raising **awareness** about inadvertent information sharing or oversharing and **educate** individuals

This will avoid the “*crisis of trust*” towards technologies

Every Day Should Be Data Protection Day



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Intentional data leakage

Cyber attacks that exploit system/service vulnerabilities or password vulnerabilities to infiltrate a data source and extract sensitive information

Users tend to use the **SAME weak** password across multiple services

Attacks based on brute force (e.g., dictionary attacks, password spraying), credential stuffing

Many devices (e.g., routers) have *default* passwords that are seldom changed by the users

Some manufacturers associate *hardcoded* (default) passwords with their IoT devices

Lack of password policy enforcement by providers/companies & lack of security awareness of users

Vulnerabilities related to software bugs and misconfigurations (e.g., APIs) are often exploited to compromise privacy

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

68

Passwords: old habits die hard

Digital safety: Are you using one of the world's most popular passwords?

Italy

Rank	Password	Time to crack it	Rank	Password	Time to crack it
1	123456	< 1 second	1	123456	< 1 second
2	123456789	< 1 second	2	ca\$h1am!	< 1 second
3	12345678	< 1 second	3	123456789	< 1 second
4	password	< 1 second	4	12345678	< 1 second
5	qwerty123	< 1 second	5	password	< 1 second
6	qwerty1	< 1 second	6	juventus	< 1 second
7	111111	< 1 second	7	qwerty1	< 1 second
8	12345	< 1 second	8	qwerty123	< 1 second
9	secret	< 1 second	9	francesco	7 seconds
10	123123	< 1 second	10	qwerty	< 1 second
11	1234567890	< 1 second	11	Abcd1234	< 1 second
			12	alessandro	1 hour

Source: NordPass

Simple, short and predictable passwords leave people highly vulnerable to data breaches

Since 2016, billions online credentials stolen

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

69

Data breaches in the media

Mother of all breaches reveals 26 billion records: what we know so far

Updated on: January 29, 2024 10:07 AM  3

The supermassive leak contains data from numerous previous breaches, comprising an astounding 12 terabytes of information, spanning over a mind-boggling 26 billion records. The leak, which contains LinkedIn, Twitter, Weibo, Tencent, and other platforms' user data, is almost certainly the largest ever discovered.

Big brands caught in 'mother of all breaches'

Twitter, LinkedIn, Adobe among set of 26 billion records.

By Leonard Bernardone on Jan 25 2024 11:48 AM

[Check if your data has been leaked](#)

BRANDS WITH 100M+ LEAKED RECORDS	
BRAND NAME	RECORDS LEAKED
Tencent	1.5B
Weibo	504M
MySpace	360M
Twitter	281M
Wattpad	271M
NetEase	261M
Deezer	258M
LinkedIn	251M
AdultFriendFinder	220M
Zynga	217M
Luxottica	206M
Evite	179M
Zing	164M
Adobe	153M
MyFitnessPal	151M
Canva	143M
JD.com	142M
Badoo	127M
VK	101M
Youku	100M

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

70

Data breaches in the media

Toyota Confirms Exposure of Customer and Employee Data in Data Breach

Toyota has confirmed that its network was breached, resulting in a 240 GB data leak.

Slack Security Breach: External GitHub Repositories Compromised, Private Source Code Taken

by Wei Chieh Lim | Jan 9, 2023

"On December 29, 2022, we were notified of suspicious activity on our GitHub account. Upon investigation, we discovered that a limited number of Slack employee tokens were stolen and misused to gain access to our externally hosted GitHub repository. Our investigation also revealed that the threat actor downloaded private code repositories on December 27. No downloaded repositories contained customer data, means to access customer data, or Slack's primary codebase."

If You Use LastPass, Change All of Your Passwords ASAP

It's time to start looking for a different password manager after the latest LastPass data breach.

 **Attila Tomaschek**
Feb. 16, 2023 9:00 a.m. PT

7 min read 

Microsoft network breached through password-spraying by Russia-state hackers

Senior execs' emails accessed in network breach that wasn't caught for 2 months.

DAN GOODIN - 1/29/2024, 2:41 AM

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

71

Pwned websites

Breached websites that have been loaded into Have I Been Pwned



Deezer

In late 2022, the music streaming service Deezer disclosed a data breach that impacted over 240M customers. The breach dated back to a mid-2019 backup exposed by a 3rd party partner which was subsequently sold and then broadly redistributed on a popular hacking forum. Impacted data included 229M unique email addresses, IP addresses, names, usernames, genders, DoBs and the geographic location of the customer.

Breach date: 22 April 2019

Date added to HIBP: 2 January 2023

Compromised accounts: 229,037,936

Compromised data: Dates of birth, Email addresses, Genders, Geographic locations, IP addresses, Names, Spoken languages, Usernames

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

72

Pwned websites

244 Million Passwords Stolen From Crime Forum —Are Yours On The List?

By [Davey Winder](#), Senior Contributor. Davey Winder is a veteran cybersecurity ...

Feb 26, 2025, 06:07am EST

Channel Info



alien

8,897 subscribers



ALIEN TXTBASE Stealer Logs

In February 2025, 23 billion rows of stealer logs were obtained from a Telegram channel known as ALIEN TXTBASE. The data contained 284M unique email addresses alongside the websites they were entered into and the passwords used. This data is now searchable in HIBP by both email domain and the domain of the target website.

Breach date: 15 February 2025

Date added to HIBP: 25 February 2025

Compromised accounts: 284,132,969

Compromised data: Email addresses, Passwords

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

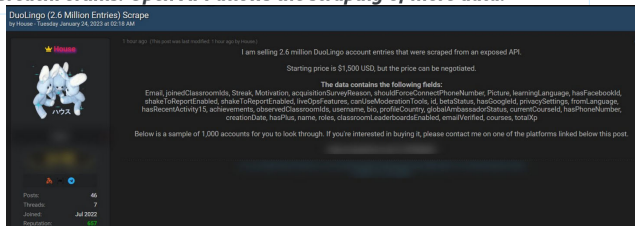
73

Data leaks due software bugs

Hackers exposed 2.6 million Duolingo users, more available for scraping

Updated on: November 15, 2023 12:53 PM

Duolingo, the popular language learning app, has had some of its users exposed online. The scraped data of 2.6 million people, which was on sale in January, is now available on the cybercrime marketplace BreachForums. Open API allows the scraping of more data.



OpenAI CEO admits a bug allowed some ChatGPT users to see others' conversation titles

PUBLISHED THU, MAR 23 2023-9:58 AM EDT | UPDATED MON, APR 17 2023-2:00 AM EDT

 Ryan Browne
@RYAN_BROWNE

© M. C. Calzarossa 2025

A Nasty Safari Bug Could Leak Your Browsing Data And Google Account Information

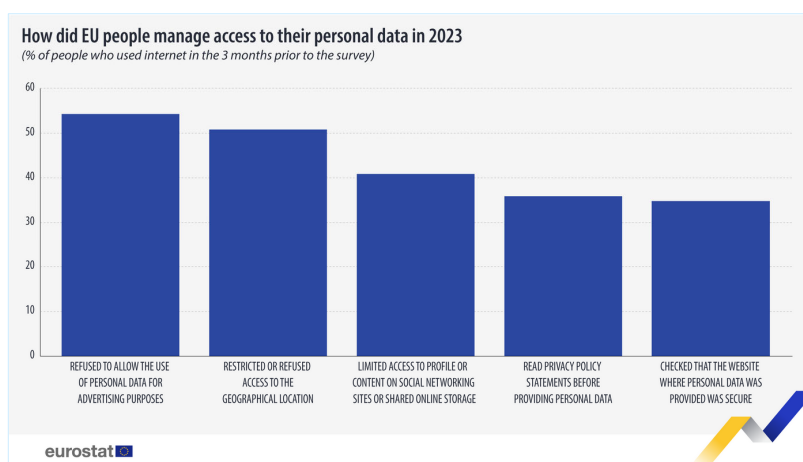
By [Lee Mathews](#), Former Contributor. Observing, pondering, and writing about tech. Generally in that order.

OpenAI CEO Sam Altman said the company fixed a vulnerability in its AI chatbot ChatGPT that meant some users were able to see the titles of chats had by others.

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Users and personal data: Eurostat survey

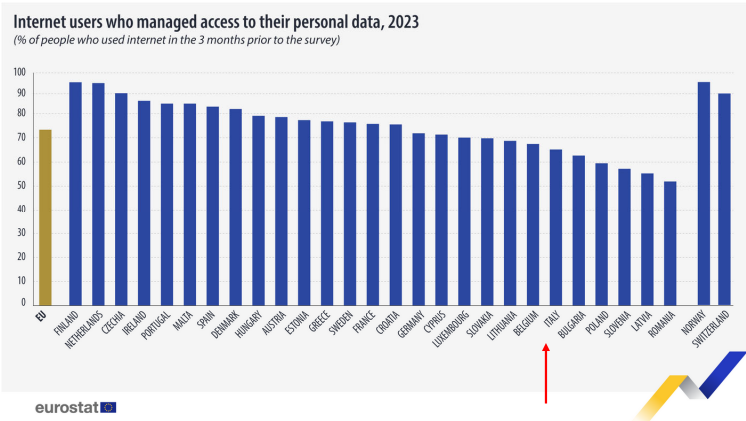


Only 36% of EU internet users read privacy policy statements before providing personal data

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Eurostat survey (cont'd)



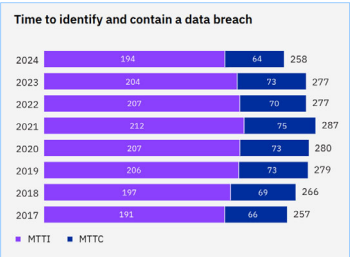
Nearly three quarters (73%) of EU Internet users managed the access to their personal data online in some way

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

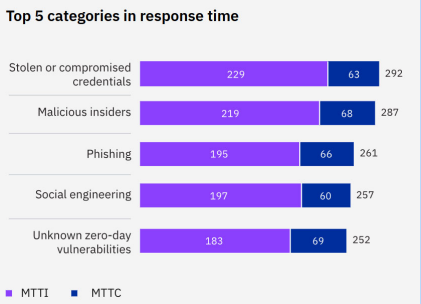
76

Time to identify/contain a data breach



Measured in days

Source: Ponemon Institute



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Measured in days

77

Type of data compromised



Source: Ponemon Institute

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

78

Cost of a data breach by country

#	Country	2024	2023
1	United States	\$9.36	\$9.48
2	Middle East	\$8.75	\$8.07
3	Benelux	\$5.90	—
4	Germany	\$5.31	\$4.67
5	Italy	\$4.73	\$3.86
6	Canada	\$4.66	\$5.13
7	United Kingdom	\$4.53	\$4.21
8	Japan	\$4.19	\$4.52
9	France	\$4.17	\$4.08
10	Latin America	\$4.16	\$3.69
11	South Korea	\$3.62	\$3.48
12	ASEAN	\$3.23	\$3.05
13	Australia	\$2.78	\$2.70
14	South Africa	\$2.78	\$2.79
15	India	\$2.35	\$2.18
16	Brazil	\$1.36	\$1.22

Measured in million USD

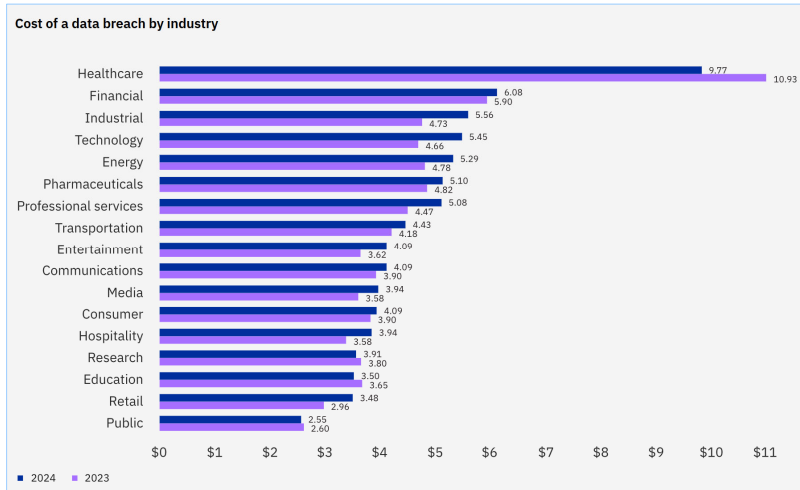
Source: Ponemon Institute

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

79

Cost of a data breach by vertical

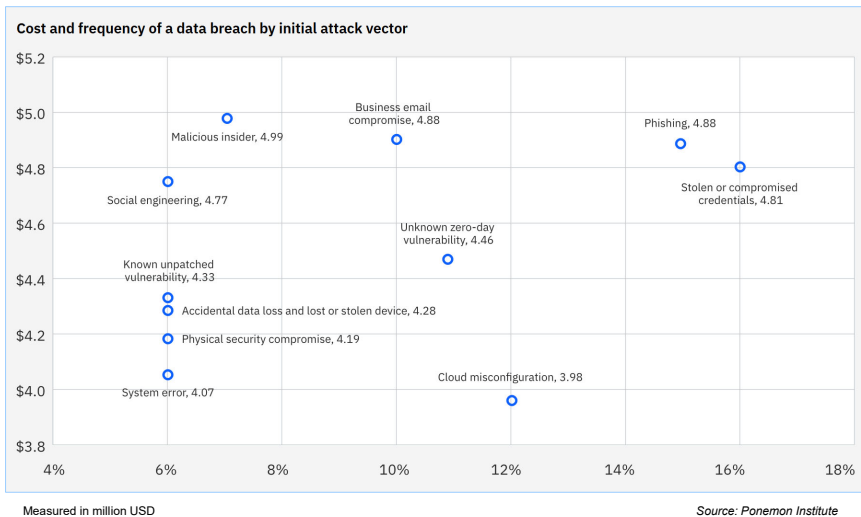


© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

80

Cost & frequency by attack vector



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

81

Recap

Users (we) are responsible of unintended data leakages because of a general tendency of oversharing

Intentional data leakages are the result of security attacks aimed at compromising user accounts and infiltrating systems and services

Users (we) are often responsible of making these attacks easier (e.g., weak passwords)

Data leakages are very costly for companies and organizations and also for individuals

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

82

Privacy: Recommendations (for everybody)

- Do not use *weak* passwords
- Do not use the *same* password for multiple services
- Do not leave *default* passwords on any device
- Do not *overshare* information online
- Be careful when *opening* URLs or attachments inside email messages
- Be careful with *camera/microphone*-equipped devices in your home/office
- Reduce your surface attack area
- Do not use *obsolete* operating systems and applications

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

83

How to prevent security threats?

A fundamental step refers to security risk assessment to identify, evaluate and prioritize potential threats and vulnerabilities, mitigate risks and enhance security measures

The preventive measures depend on the security requirements of services and infrastructure to be protected (e.g., critical assets)

Preventive security measures include:

- Ensuring all software and devices have security *patches* applied regularly
- Ensuring the network is securely *configured*
- Performing regular *data backups*, stored independently and/or appropriately protected
- Developing policies to *manage* and *monitor* access to sensitive data, including audit trails for changes to records
- Understanding the *reliance* on digital infrastructure and have a business continuity plan

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

84

Encryption

Encryption is the process of converting readable data (*plaintext*) into an unreadable format (*ciphertext*) using a mathematical algorithm and a secret key

Only authorized parties with the correct decryption key can reverse the process and read the original data

Encryption is strong measure used to ensure data confidentiality and integrity

Encryption can be applied to

- *Data in transit*, i.e., transferred over networks
- *Data at rest*, i.e., inactive data stored on a device

Encrypted data can be safely stored in the cloud or on mobile devices, thus ensuring data remains accessible anywhere, anytime

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

85

Access control

Access control mechanisms refer to the techniques and processes used to control and manage access to data and resources within a system

Their goal is to ensure security and integrity of sensitive data and information, and mitigate the risks of unauthorized accesses and data breaches

Implemented by restricting the access to data/resources to “authorized” users by specifying who can access what resources, under what conditions, and with what level of privileges

Authentication is an *explicit* access control mechanism aimed at verifying the identity of a user (e.g., credentials, pins, biometric data)

Authorization is an *implicit* mechanism that determines the actions and resources of an authenticated user by granting or denying access permissions based on the user identity and defined roles; authorization also based on device properties (e.g., MAC address, IP address)

Sophisticated mechanisms, e.g., Role-Based Access Control, Access Control Lists, for managing access permissions

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

86

Access control: best practices

Implementing effective authentication and authorization mechanisms is crucial for maintaining the security and integrity of a system

Best practices include:

- Password enforcement mechanisms
- Two-factor authentication (2FA) with One Time Password (OTP)
- Regularly reviewing and updating user roles and access permissions
- Enforcing the principle of least privilege, granting users only the necessary access permissions

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

87

Yet another security measure

Users are often the weakest link of the security chain

To ensure security and privacy a very important measure is

User education

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

88

General Data Protection Regulation



[General Data Protection Regulation](#) (GDPR) is a general framework for data protection laws in the European Union effective since May 25, 2018

It provides rules on the protection of natural persons with regard to processing personal data (PII) and on the free movement of such data

It applies to EU citizens and to any company that processes their data

Data controller determines the purposes and means of processing personal data, that is, how and why data is processed

Data processor (an employee or a third party) processes the data on behalf of the data controller

Rules refer to the obligations of data controllers/processors (e.g., consent to store/transfer/process data, rights to data access, data breach notifications)

Consent must be “*freely given, specific, informed and unambiguous*”

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

89

GDPR compliance

GDPR compliance is a *legal* requirement: data controllers must ensure that the GDPR rules are strictly followed by their services

Companies that violate GDPR privacy and security standards risk severe penalties, including fines of up to €20 million or 4% of their annual revenue, whichever is higher

451: Unavailable due to legal reasons

We recognize you are attempting to access this website from a country belonging to the European Economic Area (EEA) including the EU which enforces the [General Data Protection Regulation](#) (GDPR) and therefore access cannot be granted at this time. For any issues, contact webmaster@unicondemocrat.com or call 209-532-7191.

To check and make a service GDPR compliant, it is necessary to:

- Perform a *GDPR impact assessment* (using specific software tools) to ensure data controllers properly address privacy and data protection risks (including those associated with third-party services)
- Provide detailed information about which data are being collected and how they are used, shared, ...
- Implement mechanisms to obtain the user consent
- Implement mechanisms to allow users to access/delete/correct their data
- Protect data being collected/processed from unauthorized access and misuse with appropriate data security measures

© M. C. Calzarossa 2025

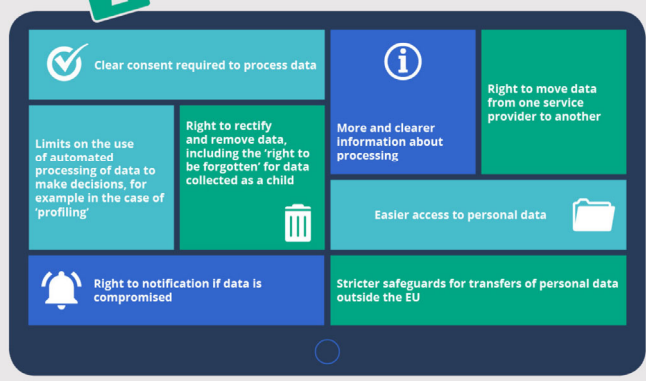
Enterprise Digital Infrastructure

90

European data protection for the digital era



Better protection for personal data



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

91

What media say about GDPR violations



Italy fines OpenAI over ChatGPT privacy rules breach

By Eivira Pollina and Alvise Armellini
December 20, 2024 9:51 PM GMT+1 · Updated 3 months ago

Italy's privacy watchdog fines OpenAI €15 million after probe into ChatGPT data collection

Company says decision disproportionate, to file appeal

The fine comes after the authority found OpenAI processed users' personal data to "train ChatGPT without having an adequate legal basis and violated the principle of transparency and the related information obligations towards users".

Italy's Garante fines Enel Energia over €79m for data privacy violations in telemarketing

A record fine of over €79m (\$85.4m) has been imposed on [Enel Energia](#) over alleged mishandling of numerous electricity and gas supply customers' personal data for telemarketing purposes.

Italy watchdog fines Foodinho 5 million euros for rider data breaches

By Reuters

November 22, 2024 9:24 PM GMT+1 · Updated 4 months ago

The regulator also prohibited Foodinho - a unit of Spanish start-up Glovo which Delivery Hero ([DHER.DE](#)) bought in 2022 - from using biometric data of its riders, such as facial recognition for identity verification.

Uber fined in Netherlands for sending drivers' data to the US

By Reuters

August 26, 2024 8:09 PM GMT+2 · Updated 6 months ago



Facebook parent company fined €91m over password storage

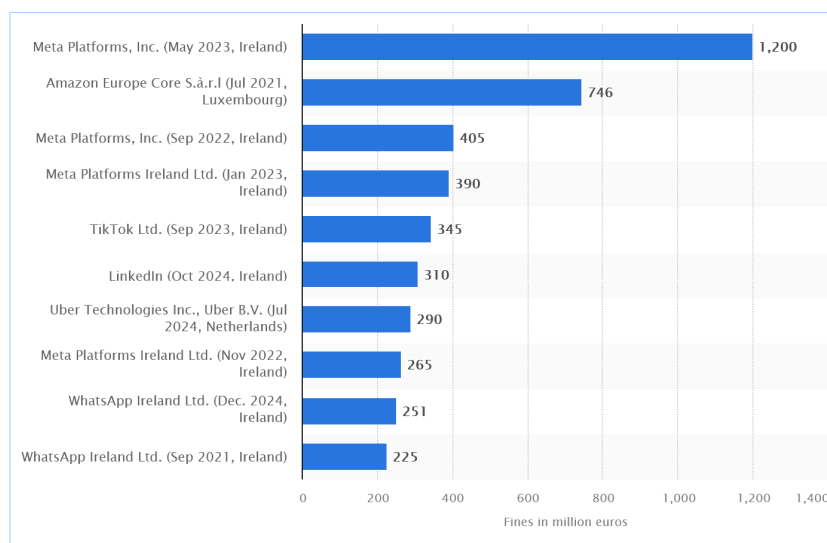
An inquiry was launched in April 2019 after Meta notified the DPC that it had inadvertently stored certain passwords of social media users on its internal systems without encryption.

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

92

Largest fines for GDPR violations



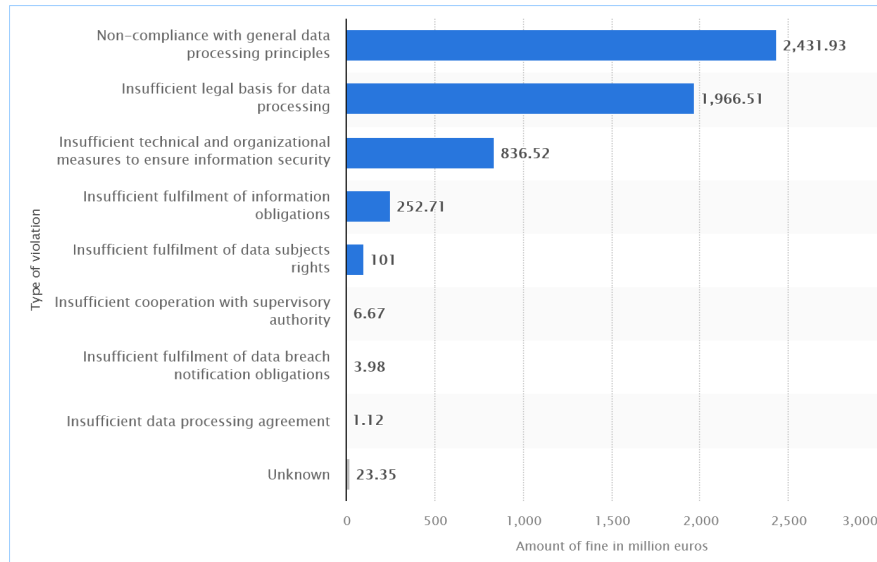
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

© Statista 2025

93

Fines by violation type



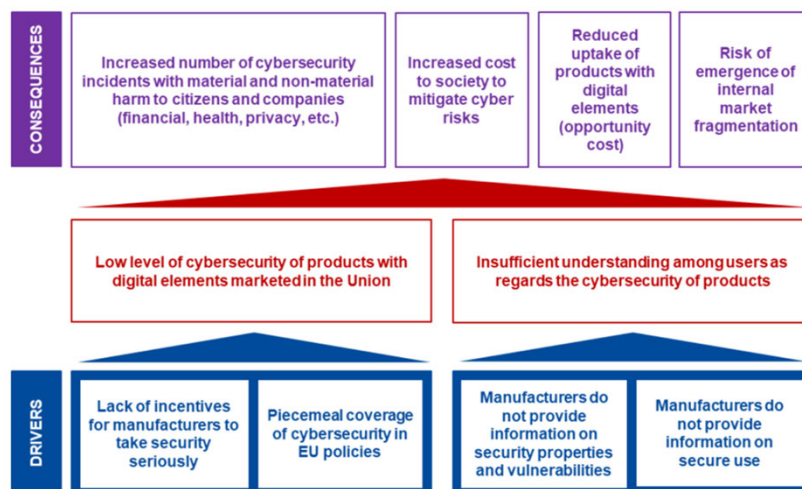
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

© Statista 2025

94

Current scenario on cybersecurity



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

95

Recap

Different measure to prevent security incidents and ensure privacy

Encryption is a strong measures aimed at protecting data in transit and at rest

Authentication and authorization mechanisms protect resources and data by associating access permissions that depend on the user identity and role

The protection of personally identifiable data is also stated by the GDPR that introduces a set of rules companies and organizations have to comply with

European legislation on cyber security

Main motivations:

- The number of discovered vulnerabilities (CVEs) is increasing
- The large majority of CVEs is exploitable over the Internet
- No incentives to develop hardware and software *secure by design*



EU Cybersecurity Act

The [*Cybersecurity Act*](#) aims at improving protection against cybersecurity threats within the European Union [adopted in 2019]

It establishes a EU-wide cybersecurity *certification framework* for ICT products, services and processes

The framework consists of a comprehensive set of rules, technical requirements, standards and procedures

Certification managed by ENISA (European Union Agency for Cybersecurity)

Companies doing business in the EU will be able to demonstrate that they have adopted security best practices, thus increasing stakeholder trust

Certificates are recognized across the entire EU

Companies need to certify their ICT products, processes and services only once

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

98

EU Cyber Resilience Act



The [*Cyber Resilience Act*](#) (CRA) is the first EU legislative framework placing *mandatory* cybersecurity requirements for *products that include digital elements*

CRA entered in force on 10 December 2024, and its main obligations will apply from 11 December 2027

Products with digital elements: any software or hardware product and its remote data processing solutions, including software or hardware components to be placed on the market separately

CRA aims at protecting consumers and businesses from hardware and software products with inadequate security features

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

99

EU Cyber Resilience Act (cont'd)

The design, development and production of products with digital elements have to ensure their conformity with *mandatory* security requirements dependent on their cybersecurity risks

Important products (e.g., password managers, operating systems, virtual assistants) require a third-party assessment

Critical products (e.g., smart meters, smart cards) require a certification

Manufacturers must guarantee the security of hardware and software products throughout their *whole lifecycle* by providing software updates that fix security vulnerabilities and offering security support to consumers (for free)

These rules will benefit consumers and businesses using digital products by enhancing the transparency of the security properties and promoting trust

Products will bear the CE marking to indicate that they comply with the regulation requirements



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

100

Impacts of Cybersecurity

Security must be integrated in the entire lifecycle of digital products to comply with legislation and avoid incidents due to vulnerabilities (*security by design*)

Direct costs of security refer to the investments in terms of human resources and technological infrastructure related to the design and development of “secure” products and the prevention/detection/mitigation of security threats

Indirect costs of security refer to the costs related to business and reputation losses due to data breaches and security incidents

Setup of an appropriate *incident response plan*, i.e., a set of instructions with the specification of the role and responsibility of the security team, to help security team detect, respond to, and recover from security incidents

All these requirements increase the overall cost associated with services/infrastructures

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

101

Dependability: summary

Availability: readiness

- » scheduled/unscheduled downtimes affect availability
- » downtimes are due to maintenance activities, human errors, bugs, misconfigurations, security attacks & reliability of technological infrastructure

Reliability: continuity

- » robustness of components
- » appropriate redundancy, replication, distribution & diversity

Trustworthiness:

- » security & privacy, confidentiality, integrity & availability
- » security threats affect infrastructure/services and data
- » measures to protect infrastructure/services, prevent security incidents and promptly react in case incidents happen
- » measures to educate users & create awareness towards security principles

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

102

Performance

Performance is a quality attribute that measures the *efficiency* of a service or technological infrastructure in terms of *speed* and *time* to complete a given task

Performance is often considered as a synonym of quality

Very subjective attribute

Performance affects usability

Two metrics are used to measure performance:

- *Throughput*, i.e., amount of work performed per time unit (e.g., bandwidth, flops, fps)
 - it should be **maximum**
- *Time*, i.e., how long it takes to complete a given task (e.g., network latency, page load time, response time)
 - it should be **minimum**

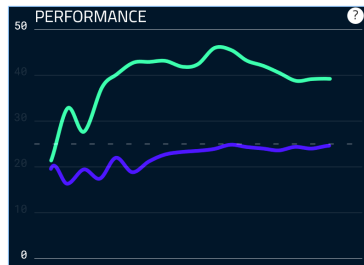
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

103

Performance (cont'd)

It is difficult to ensure “good” performance levels



Actual performance affected by the characteristics and intensity of the workload/traffic (e.g., nominal vs actual bandwidth, on demand services, social nature of services, virtualized resources)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

104

Service Level Agreement

A Service Level Agreement is a **contract** between an external or internal provider and its customers

The contract is a *legally binding* document that specifies *technical*, *legal* and *administrative/financial* provisions about the service

These *provisions* refer to:

- *conditions* under which the service is delivered by the provider
- *responsibilities* (i.e., rights, duties) of the customers
- *service levels* to be expected

Legal Stuff

The contract must ensure that both parties have the same understanding of the agreed provisions

No party can plead ignorance!



Contracts vary with the type of service (e.g., premium, free), type of customers (e.g., business, consumer) and cost associated with the service

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

105

SLA Provisions

The contract is a *live document* that includes:

- Description of the *service* (i.e., what service is about and what provider offers)
- Description of the *service levels* (i.e., how provider will implement the offer, expressed in terms of *quantitative/measurable* quality indicators)
- Description of the service level *monitoring & reporting* (i.e., who/how/how often the indicators will be measured and made available to customers)
- Description of consequences for *violations* of service obligations (i.e., penalties, indemnifications, contract termination); force-majeur excluded
- Description of *customer responsibilities* (i.e., what customers can do or must not do)
- Description of *contract evolution* (i.e., renewal, duration, changes)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

106

Example of SLA template



Table of Contents

1.0 Service Level Agreement	2
1.1 Version Details	2
1.2 Document Change History	2
1.3 Document Approvals	2
2.0. Agreement Overview	2
2.1. SLA Introduction	2
2.2. Definitions, Conventions, Acronyms and Abbreviations	2
2.3. Purpose	3
2.4. Contractual Parameters	3
3.0. Service Agreement	3
3.1. KPIs and metrics	3
3.2. Service levels, rankings and priority	3
3.3. Service Response	4
3.4. Exceptions and Limitations	4
3.5. Responses and responsibilities	4
3.6. Service Management	5
References and Glossary	5
Appendix	5
A.1. Pricing models and charges	5

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

107

Google Terms of Service



Effective May 22, 2024 | [Archived versions](#) | [Download PDF](#)

Country version: [Italy](#)

What's covered in these terms

We know it's tempting to skip these Terms of Service, but it's important to establish what you can expect from us as you use Google services, and what we expect from you.

[Google Terms of Service](#)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

These Terms of Service reflect [the way Google's business works](#), the laws that apply to our company, and [certain things we've always believed to be true](#). As a result, these Terms of Service help define Google's relationship with you as you interact with our services. For example, these terms include the following topic headings:

- [What you can expect from us](#), which describes how we provide and develop our services
- [What we expect from you](#), which establishes certain rules for using our services
- [Content in Google services](#), which describes the intellectual property rights to the content you find in our services — whether that content belongs to you, Google, or others
- [In case of problems or disagreements](#), which describes other legal rights you have, and what to expect in case someone violates these terms

Understanding these terms is important because, to use our services, you must accept these terms.

Besides these terms, we also publish a [Privacy Policy](#). Although it's not part of these terms, we encourage you to read it to better understand how you can [update, manage, export, and delete your information](#).

108

Google Privacy Policy



GOOGLE PRIVACY POLICY

When you use our services, you're trusting us with your information. We understand this is a big responsibility and work hard to protect your information and put you in control.

This Privacy Policy is meant to help you understand what information we collect, why we collect it, and how you can update, manage, export, and delete your information.

If European Union or United Kingdom data protection law applies to the processing of your information, you can review the [European requirements section](#) below to learn more about your rights and Google's compliance with these laws.



Privacy Checkup

Looking to change your privacy settings?

[Take the Privacy Checkup](#)

Effective September 16, 2024 | [Archived versions](#) | [Download PDF](#)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Google Privacy & Terms	
Overview	Privacy Policy
Introduction	
Information Google collects	
Why Google collects data	
Your privacy controls	
Sharing your information	
Keeping your information secure	
Exporting & deleting your information	
Retaining your information	
Compliance & cooperation with regulators	
European requirements	
About this policy	
Related privacy practices	

[Google Privacy Policy](#)

109

AMAZON ELASTIC COMPUTE CLOUD (EC2) (INSTANCE LEVEL)

Instance-Level SLA

For each individual Amazon EC2 instance ("Single EC2 Instance"), AWS will use commercially reasonable efforts to make the Single EC2 Instance available with an Instance-Level Uptime Percentage of at least 99.5%, in each case during any monthly billing cycle (the "Instance-Level SLA"). In the event any Single EC2 Instance does not meet the Instance-Level SLA, you will be eligible to receive a Service Credit as described below.

Instance-Level Uptime Percentage	Service Credit Percentage
Less than 99.5% but equal to or greater than 99.0%	10%
Less than 99.0% but equal to or greater than 95.0%	30%
Less than 95.0%	100%

Note: In addition to the Instance-Level SLA, AWS will not charge you for any Single EC2 Instance that is Unavailable for more than six minutes of a clockhour. This applies automatically and you do not need to request credit for any such hour with more than six minutes of Unavailability.

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

110

Amazon S3 Service Level Agreement

Last Updated: November 28, 2023

AMAZON SIMPLE STORAGE SERVICE (S3)

Monthly Uptime Percentage	Service Credit Percentage
Less than 99.9% but greater than or equal to 99.0%	10%
Less than 99.0% but equal to or greater than 95.0%	25%
Less than 95.0%	100%

Amazon Simple Storage Service (S3) Full SLA

Serverless | Storage

Definitions

- "Error Rate" means, for each request type to an Amazon S3 storage class: (i) the total number of internal server errors returned by Amazon S3 for such request type to the applicable Amazon S3 storage class as error status "InternalError" or "ServiceUnavailable" divided by (ii) the total number of requests for such request type during the applicable 5-minute interval of the monthly billing cycle. We will calculate the Error Rate for each AWS account as a percentage for each 5-minute interval in the monthly billing cycle. The calculation of the number of internal server errors will not include errors that arise directly or indirectly as a result of any of the Amazon S3 SLA Exclusions.
- "Monthly Uptime Percentage" is calculated by subtracting from 100% the average of the Error Rates from each 5-minute interval in the monthly billing cycle. If you did not make any requests in a given 5-minute interval, that interval is assumed to have a 0% Error Rate.

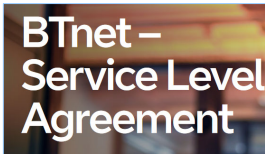
Amazon S3 SLA Exclusions

The Service Commitment does not apply to any unavailability, suspension or termination of Amazon S3, or any other Amazon S3 performance issues: (i) caused by factors outside of our reasonable control, including any force majeure event or Internet access or related problems beyond the demarcation point of Amazon S3; (ii) that result from any actions or inactions of you; (iii) that result from your equipment, software or other technology; or (iv) arising from our suspension or termination of your right to use Amazon S3 in accordance with the AWS Agreement (collectively, the "Amazon S3 SLA Exclusions"). If availability is impacted by factors other than those used in our calculation of the Monthly Uptime Percentage, then we may issue a Service Credit considering such factors at our discretion.

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

111



Network latency

1. We agree to provide your service with a latency guarantee subject to the terms of this agreement.
2. We'll work out this latency by calculating the average of round-trip transmission measurements taken in ten minute intervals during a calendar month. The results will be posted on www.bt.net.
3. The latency guarantee depends on the network you're using:
The UK core network connection: The latency guarantee is an average round-trip transmission time of 20 milliseconds or less between BT selected core Point of Presence (PoP) in the BT network.
The transatlantic network connection: The latency guarantee is an average round-trip transmission time of 95 milliseconds or less between designated transit routers at each end of the transatlantic link.
4. If we don't meet the latency guarantee for two consecutive months we'll give you a discount. You can save the equivalent to one day's rental charge for your service.
5. You don't need to make a claim, we'll just automatically apply the reduction to your rental charges.

Service availability

1. We aim to provide your service with a target of 100% availability at all times, subject to the terms of this agreement.
2. If there's evidence been an outage, we'll reduce your BTnet rental charge for the service as follows:
 - Where the outage period lasts for ten hours or less, you'll save the equivalent of one day's rental charge per hour of downtime. When calculating the outage period, we'll round up the total time to the nearest hour.
 - The maximum reduction applied per quarter will be capped at ten hours of outage, which is equivalent to ten days' rental charge for your service.
3. You should report an outage within two days by calling 0808 100 2440. We'll provide you with a trouble ticket number for your reference.
4. Once we've verified the outage, you'll need to make a claim within 28 days, quoting the relevant trouble ticket number, customer reference (FTIP number) and reason for the claim by email to: btnet.slg@bt.com

© M. C. Calzarossa 2025
Enterprise Digital Infrastructure

112

Why performance varies

Poor performance and inefficiencies are due to load/traffic conditions or inadequate infrastructures and lead to SLA violations & financial/reputation losses

Unforeseen interactions between components (that perform well in isolation) might lead to poor performance

A failure of a component might overload other components (*cascade effects*), thus leading to poor performance

It is necessary to identify precisely the causes of the performance problems

Bottleneck: the component that limits the performance of a system

A component is not a bottleneck by definition

It might become a bottleneck



© M. C. Calzarossa 2025
Enterprise Digital Infrastructure

113

Performance engineering approaches

Performance engineering is the discipline that studies the efficiency of systems/services (under design or being deployed) and the quality experienced by the users

Two types of performance engineering approaches:

- *reactive* approach whose objective is to find remedies as soon as problems appear by applying (temporary or permanent) *tuning* actions (i.e., “adjustments” of the behavior of an infrastructure to mitigate the symptoms and effects of the bottleneck)
- *proactive* approach whose objective is to predict/anticipate performance problems by applying *capacity planning* actions (i.e., analysis of the current conditions of an infrastructure, prediction of its residual capacity and of its evolution through a *what-if* analysis)

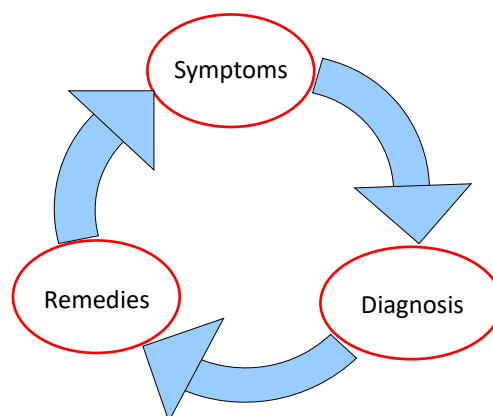
Both approaches should be applied (i.e., anticipation & reaction)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

114

How to identify/remove performance issues?



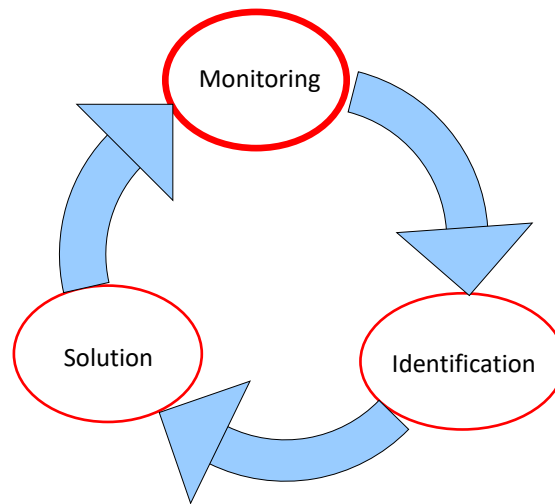
All these activities require a solid knowledge of the functionalities and behavior of systems/services (*domain knowledge*)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

115

Performance analysis in detail



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure